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Building Services and Information

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1 A. Building Environment and IEQ

P001 — Spatial-temporal CO2 concentration prediction in education facilities with dynamic layouts using BIM and multi-modal graph neural networks

Authors: Hao Liu, Kam-Ming Mark Tam, Cong Huang, Chun Yin Lai, Eric Schuldenfrei

Major Category: Building Environment and IEQ

Secondary Category: AI and Machine Learning for Buildings, BIM and Semantic Information

Keywords: Spatial-temporal prediction, CO2 concentration, Education facility, Space reconfigurability, BIM, Graph neural networks, Multi-modal deep learning

■**主な主張** BIM 由来の時変レイアウトと Voronoi ベースの CO2 センサ隣接グラフを multi-modal deep learning で統合し, 2 時間先予測で MAE 11.25 ppm を達成し, 比較対象中で次点の TGNN の 15.10 ppm より約 25% 低い MAE を示した. [1]

■**新規性** 静的レイアウトを前提とする既往の GNN 型 CO2 予測に対し, BIM から時変の間仕切り構成をテンソル化し, センサグラフ文脈と空間構成文脈を二枝の MMDL で統合した点に新規性がある.

■**位置付け** 教育施設の IAQ 予測を主目的としつつ, BIM 幾何情報を予測モデルの入力文脈として利用するため, Building Environment and IEQ を主分類, AI and Machine Learning for Buildings および BIM and Semantic Information を副分類として位置付ける.

2 B. HVAC and Building Services

3 C. Building Energy and ZEB

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5 E. BIM and Semantic Information

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9 I. AI and Machine Learning for Buildings

10 J. Retrieval, Query, LLM and Agent

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References

- [1] Hao Liu, Kam-Ming Mark Tam, Cong Huang, Chun Yin Lai, and Eric Schuldenfrei. Spatial-temporal co2 concentration prediction in education facilities with dynamic layouts using bim and multi-modal graph neural networks. *Building and Environment*, 297:114528, 2026.